What we're doing

How good was the best contacts-v1 model we had on any given date — on contact R-precision and on held-out validation loss — and how tightly do those two track each other?

Every number exists somewhere already (#67, #75, #82, #89, #108, #117, #120, #146, #160, #169), but they are scattered across issue comments, per-experiment CSVs and two W&B projects, with two different inference recipes mixed in. This experiment collects them into one dataset and three standing figures, and keeps them refreshable as new models land.

Why

Not a hypothesis-testing experiment — it is a **standing tracker**. The two things it is built to make visible, both of which prior issues assert piecemeal:

1. Progress on contact accuracy has come almost entirely from the **base model**, not from inference or post-training. 2. Validation loss predicts contact accuracy **across training generations** and stops predicting it **within a generation** (#169's finding), so the loss frontier and the accuracy frontier are not the same curve.

Results so far

The accuracy frontier moved in exactly two jumps, both from the base model, neither from inference or post-training: ~0.03 to 0.425 when #75's E8 rung finished (2026-06-21), and 0.436 to 0.534 when #117's 16-epoch bs256 run finished (2026-07-22).

Between them, five weeks of post-training and inference work moved it by +0.011 (#120's re-epoch), and #160's backtracking fine-tune moved it by -0.020.

Structure predictors on the same 554 proteins: Protenix-v2 single-seq 0.603, ESMFold 0.755, ESMFold2 0.786, Protenix-v2 + MSA 0.812. We are still below all of them.

Loss tracks accuracy across generations, not inside one

The 0.053-nat #75 to #117 gap buys +0.109 R-precision — about 2 R-precision per nat. The 0.008-nat gap between #117's early-stop and final checkpoints buys nothing, and #146's 3B is 0.0012 better on loss and 0.023 worse on R-precision.

So the loss frontier is useful for deciding which checkpoints are worth scoring, and useless for picking between two checkpoints of one run.

The trap in these numbers

The same weights score ~0.086 higher under exp82's rollout recipe than under exp89's original pairwise scorer (#61/#75 E8: 0.339 to 0.425; #120: 0.350 to 0.436). That is comparable to two generations of model progress.

The figures keep the two recipes visually separate. Anything from the eval-checkpoint skill is pairwise; anything from exp82's rollout workers or the exp169 dispatcher is rollout. Never infer the recipe from the magnitude.

Open gap

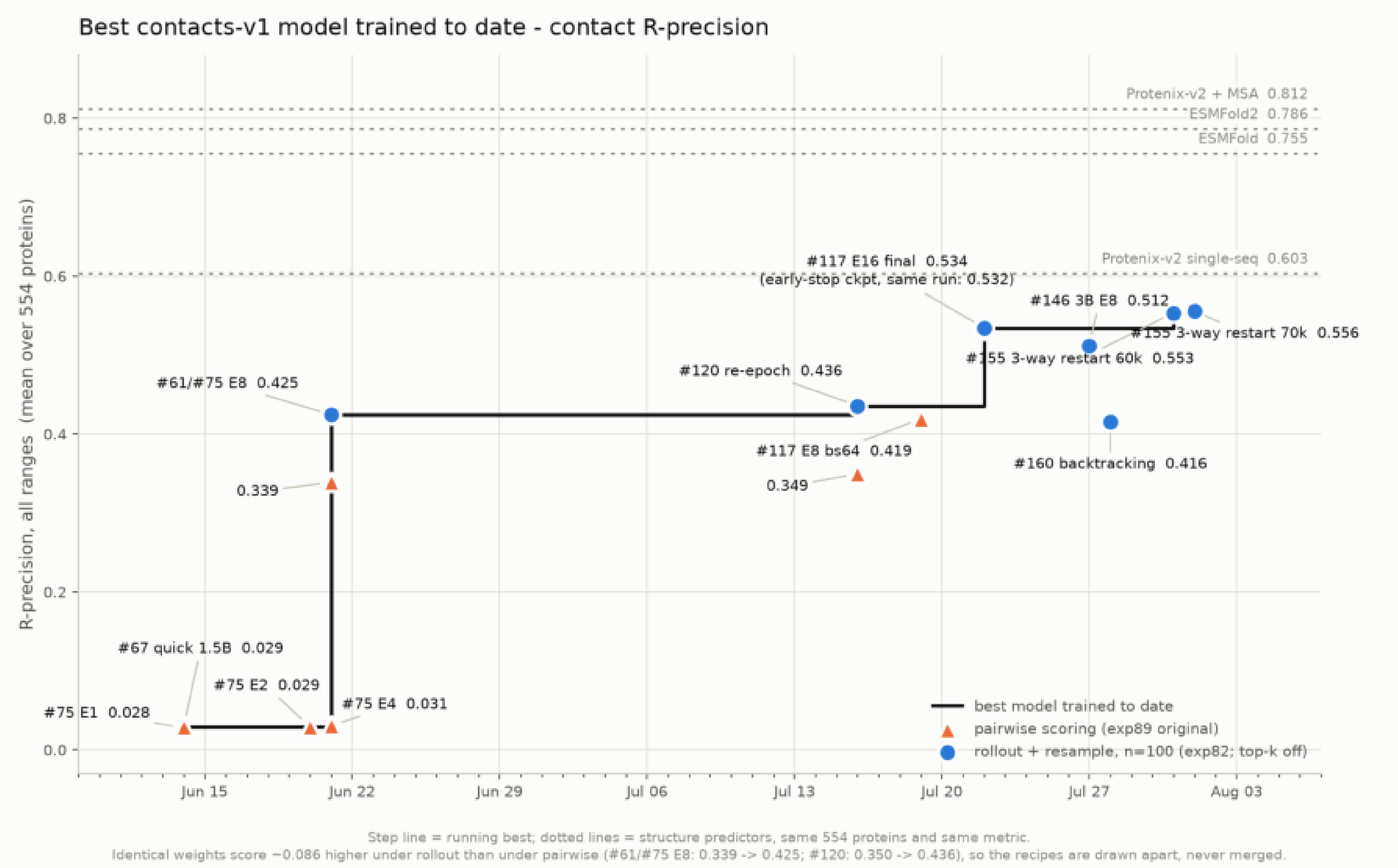
#108's 3B on CoreWeave held the loss frontier from 2026-07-11 to 07-16 at 2.7418 and was never contact-scored. Given #146 — a 3B at matched loss under-performing the 1.5B — it is probably not a missed frontier point, but that is an inference, not a measurement.

Keeping this current

Two commands: build_dataset.py re-pulls W&B, plot_progress.py redraws. New benchmark scores are hand-added to RPRECISION_ROWS with a source citation and an explicit inference recipe. Full procedure in the README.

rprecision_frontier.png

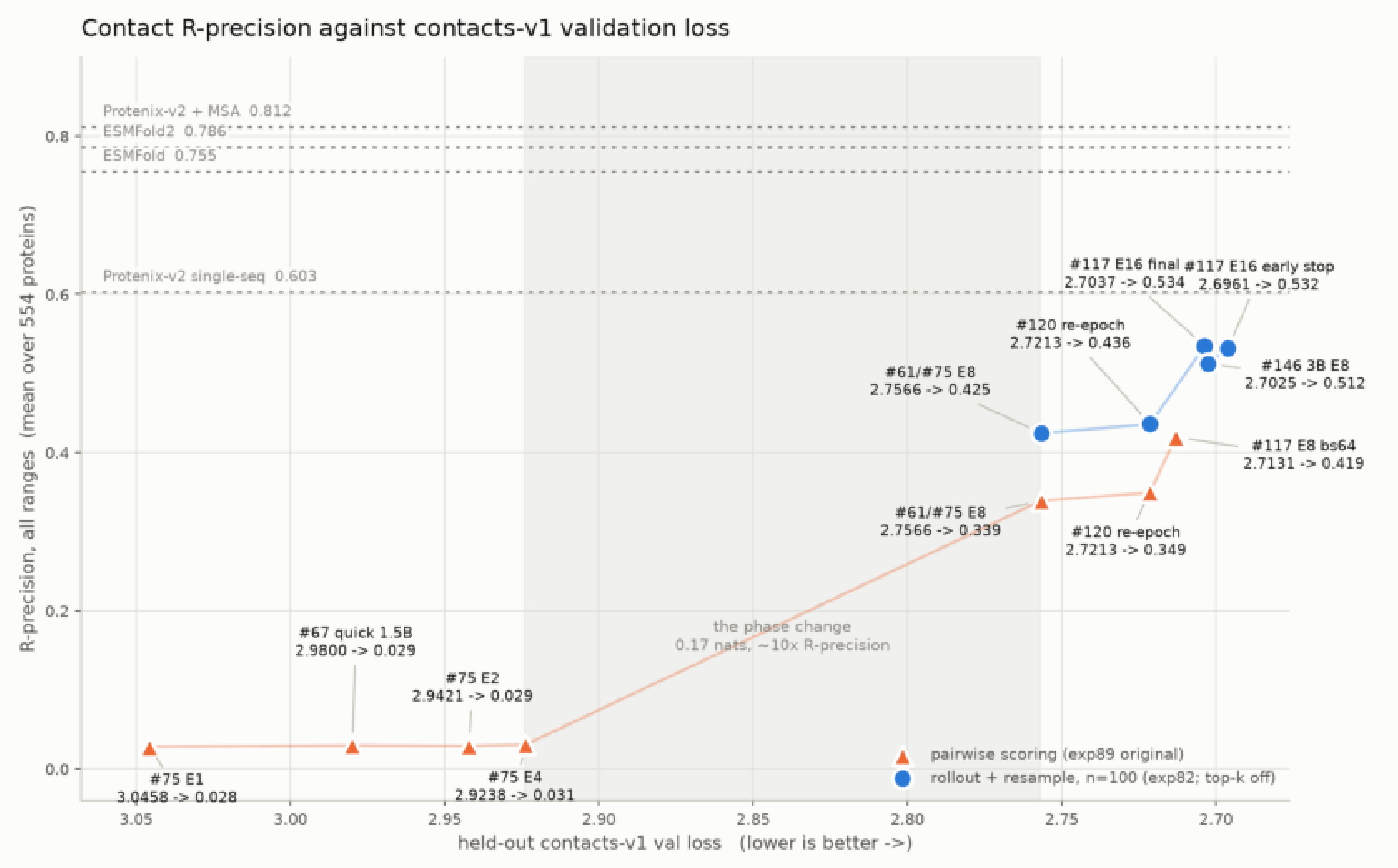
Best contacts-v1 model to date on contact R-precision. Two jumps, both from the base model: #75 E8 (Jun 21) and #117 E16 (Jul 22).



$ plot_progress.py

rprecision_vs_val_loss.png

R-precision against val loss. ~2 R-precision per nat across generations; flat inside one run (#117 early vs final).

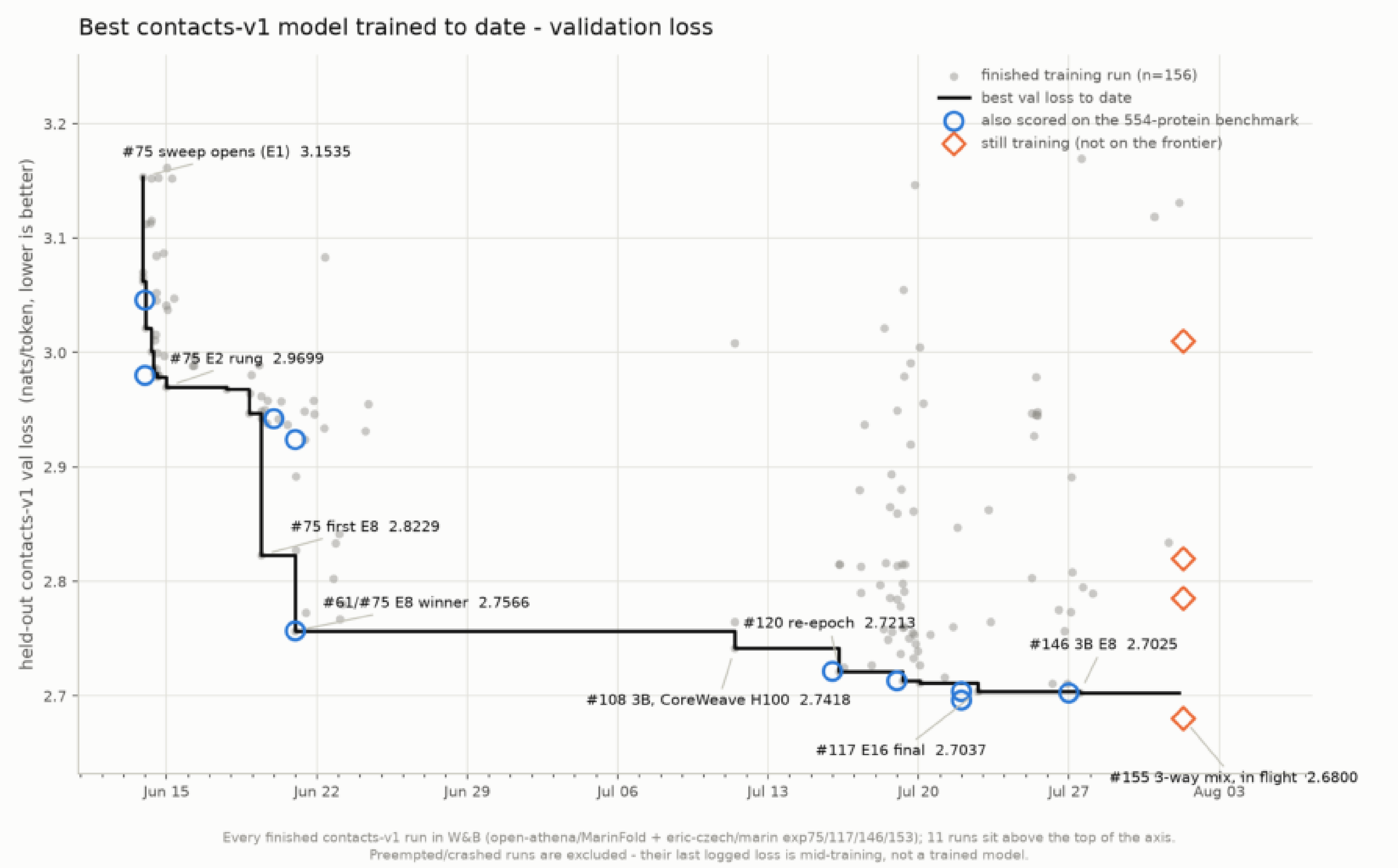


Dotted lines = structure predictors, same 554 proteins and metric; they have no LM validation loss, so they can only be horizontal here.
#120's loss is `contacts-v1-val-orig`, the others `contacts-v1-val` - the same held-out split under different cache names (see README.md).

$ plot_progress.py

val_loss_frontier.png

Best held-out contacts-v1 val loss to date over 153 finished runs. The loss frontier has many more steps than the accuracy one.



$ plot_progress.py